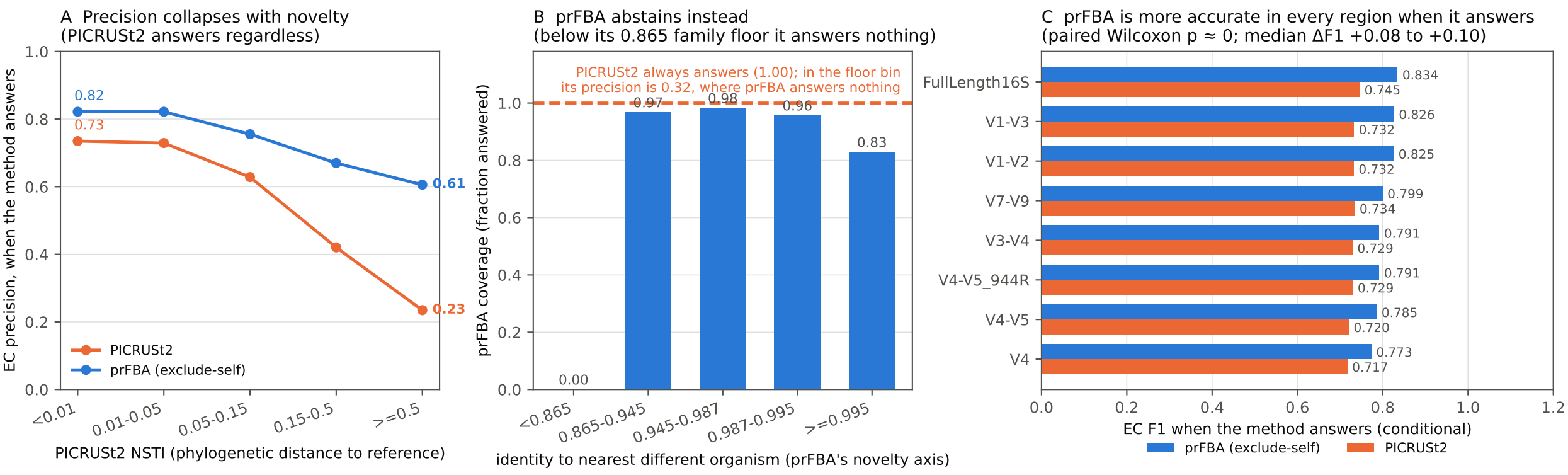
prFBA vs PICRUSt2 — EC-space functional prediction on 67,038 benchmark amplicons (10,000 genomes × 8 16S regions)
A Precision collapses with novelty
(PICRUSt2 answers regardless)
EC precision, when the method answers
1.0
0.8
0.6
0.4
0.2
0.0
0.82
0.73
0.61
0.23
PICRUSt2
prFBA (exclude-self)
<0.01
0.01-0.05
0.05-0.15
0.15-0.5
>=0.5
PICRUSt2 NSTI (phylogenetic distance to reference)
B prFBA abstains instead
(below its 0.865 family floor it answers nothing)
PICRUSt2 always answers (1.00); in the floor bin
its precision is 0.32, where prFBA answers nothing
prFBA coverage (fraction answered)
1.0
0.8
0.6
0.4
0.2
0.0
0.00
0.97
0.98
0.96
0.83
<0.865
0.865-0.945
0.945-0.987
0.987-0.995
>=0.995
identity to nearest different organism (prFBA's novelty axis)
C prFBA is more accurate in every region when it answers
(paired Wilcoxon p ≈ 0; median ΔF1 +0.08 to +0.10)
FullLength16S
0.834
0.745
V1-V3
0.826
0.732
V1-V2
0.825
0.732
V7-V9
0.799
0.734
V3-V4
0.791
0.729
V4-V5_944R
0.791
0.729
V4-V5
0.785
0.720
V4
0.773
0.717
0.0
0.2
0.4
0.6
0.8
1.0
1.2
EC F1 when the method answers (conditional)
prFBA (exclude-self)
PICRUSt2